

Fiona C. Dubay

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Education

Sarah Lawrence College

B.A. in Liberal Arts | *Cumulative GPA 3.93*

Concentration: Environmental Science

Academic Advisor: Drew Cressman | **Research Advisor:** Bernice Rosenzweig

Bronxville, NY

May 2024

Honors & Awards

Sarah Lawrence College Edward Cogan Prize for Science and Mathematics

2024

Sarah Lawrence College Scholarships:

Gaylord Donnelley Family Science Scholarship

2020-2024

Dean's Scholarship

2020-2024

Research Experience

NYC's Climate Vulnerability, Impact, and Adaptation Analysis (VIA)

New York City, NY

Student Intern

September 2023 - December 2023

- Created digital maps and visualizations using QGIS software for publication on flood exposure and risk in New York City
- Processed spatial data and databases in vector and raster formats for clear visualization in QGIS
- Communicated and collaborated with other scientists and specialists on the task team
- Assessed the quality of 311 data for analysis and visualization usage

NYC's Panel on Climate Change (NPCC) & VIA

New York City, NY

Summer Fellow

June 2023 - August 2023

- Crafted visualizations on flood exposure in NYC for publication in the NPCC's first chapter, specifically focused on flooding
- Sorted and cleaned open-source data from NYC OpenData's site for analysis use
- Handled and refined large, detailed, spatial data files through QGIS
- Utilized Python for cleaning, refining, and visualizing rainfall and tide gauge datasets

Sarah Lawrence College Summer Research Program

New York City, NY

Environmental Science Intern

2021, 2022

Advisor: Bernice Rosenzweig

- Engaged in hands-on research experience through an intensive 8-week summer program
- Assisted faculty with existing research work
- Gained a robust understanding of the means of conducting research through individual and collaborative projects
- Designed and carried out experiences, analyzed results, communicated findings with a wide audience through presentations

Sarah Lawrence College Conference Work

Bronxville, NY

Various research projects completed complementary to coursework:

2020 - 2024

- [Unequal Ecological Exchange between India and the U.K. - An analysis using WIOT Data](#)
- [Immigrants in NYC and Flood Exposure: An Investigative Data Exploration](#)
- [Terrain analysis and urban watersheds: Delineating the Upper Bronx River Basin](#)
- [Case Study: Vinyl Chloride Train Derailment in East Palestine, Ohio](#)
- [Collective Intelligence: Ant Colony Simulation & the Traveling Salesman Problem](#)
- [Radioecology of Chernobyl: The biological effects of ionizing radiation on natural systems and non-human biota](#)
- [Flood Vulnerability and 311 Service Requests: A Geospatial Analysis of Ida's Impacts on New York City Communities](#)
- [Visualizing the Microbial Communities of the Saw Mill River: an Educational Narrative](#)
- [Development in Bhutan: Happiness, Hydropolitics and India](#)
- [Lotka-Volterra Modeling: The Predator-Prey Dynamics of *Lynx canadensis* and *Lepus americanus*](#)
- [Habituation and other complex processes in *Physarum polycephalum*: what a single-celled organism can teach us about how we learn](#)
- [Declining Monarch Populations and Pesticide Use: A Data Exploration Using Python](#)

Research Interests

- Fate, transport, and remediation of environmental contaminants
- Hydrology & hydrological modeling
- Applications of remote sensing and environmental modeling

Work Experience

City of Santa Monica Office of Sustainability & the Environment

Santa Monica, CA

CivicSpark / AmeriCorps Fellow

October 2025 - Present

- Assist with community planning projects, such as developing an extreme heat risk action plan
- Mobilize volunteers and gather community feedback to support project implementation
- Gather, analyze, and organize data for various analyses and visualizations that support ongoing projects

CivilGrid

Remote

Data Research Specialist

October 2024 - February 2025

- Interpreted maps, as-built drawings, and engineering plans at a basic level
- Researched utility operations, public works assets, and asset management plans
- Digitized utility maps in ArcGIS Pro
- Communicated directly with public works offices and utility asset owners over email and phone

Agerpoint

Remote

Geospatial Data Intern

May 2024 - August 2024

- Ingested and processed numerous data types based on Standard Operating Procedures (SOPs)
- Assisted in machine learning model training by providing feedback to machine learning/AI development teams on model performance
- Organized and maintained data storage according to client account requirements and standards

Additional Experience

Starbucks Barista/Partner	May 2025 - Oct 2025
Sarah Lawrence Summer Research Program Environmental Science Intern	2021, 2022
Sarah Lawrence College SciMath Dept. Environmental Data & Microbiology Tutor	Jan 2024 - May 2024
Sarah Lawrence College IT Department IT Help Desk Assistant	Jan 2024 - May 2024
Sarah Lawrence College Library Shift Supervisor & Assistant	September 2022 - May 2023
Sarah Lawrence College Campus Van Driver	Intermittent 2021-2024
Annie Bloom's Bookstore Clerk	Seasonal 2017-2021

Skills

Technical Skills

Geospatial Analysis: QGIS, ArcGIS Pro
Data Analysis: Python (NumPy, Pandas, geopandas), R
Visualization: Python (matplotlib, seaborn), QGIS, ArcGIS Pro
Additional: CPR & First Aid Certification

Nontechnical Skills

- Independent problem solving
- Confident oral/written communication
- Organization of data
- Creativity, innovation, and adaptation

References

Bernice Rosenzweig

Sarah Lawrence College, The O'Silas Endowed Professorship in Environmental Studies

Relationship: Research Advisor

brosenzweig@sarahlawrence.edu

Drew Cressman

Sarah Lawrence College, Department Chair, The Sara Yates Exley Chair in Teaching Excellence

Relationship: Don / Academic Advisor

Tel: 914-395-2434

dcressma@sarahlawrence.edu

Michelle Hersh

Sarah Lawrence College, Professor of Biology and Ecology

Relationship: Professor, TA

mhersh@sarahlawrence.edu

Nico Predock

Sustainability Analyst - City of Santa Monica

Relationship: CivicSpark / AmeriCorps Site Supervisor

nico.predock@santamonica.gov

Vince Castillo

Store Manager - Starbucks

Relationship: Manager

Tel: 210-313-8086

Volunteer Experience

MESD Outdoor School

Student Leader / Environmental Educator

2017-2019

600 hours